

MAT602

Biomedical Materials for Genicular Artery Embolisation to Treat Pain from Knee Osteoarthritis

Professor Andrew Lewis

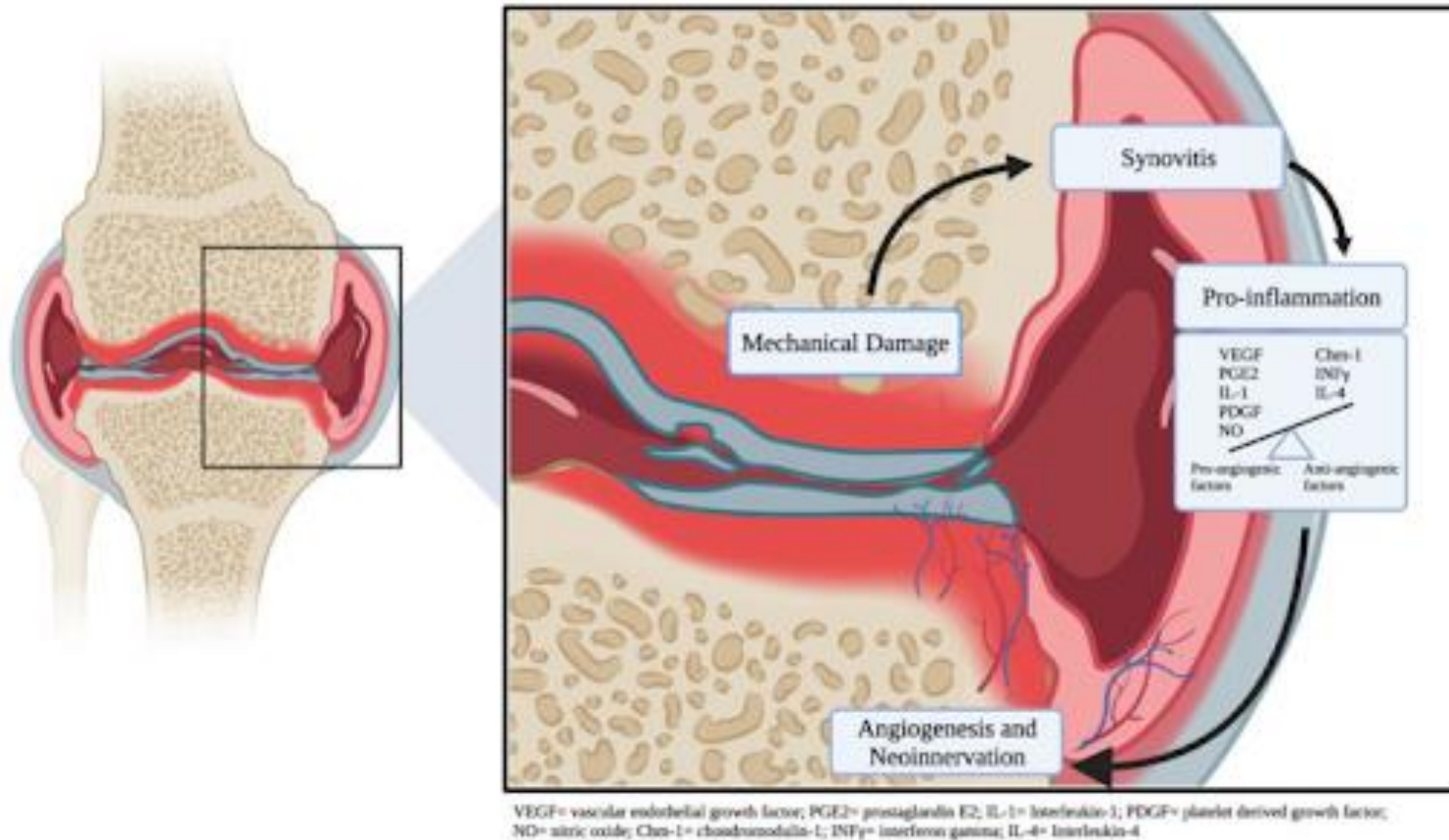
RAEng Visiting Professor, April 2025

Knee Osteoarthritis (KOA)

- Knee osteoarthritis (KOA) is a common type of arthritis that occurs when the cartilage in the knee joint wears down over time
- The global prevalence of KOA is 16% for people aged 15 and older, and 22.9% for people aged 40 and older. In the UK, an estimated 5.4 million people have KOA
- KOA is a long-term condition that can't be cured, but there are many ways to treat its symptoms. Pain from KOA can be severely debilitating and common treatments such as anti-inflammatory medication and steroid injections have limited efficacy



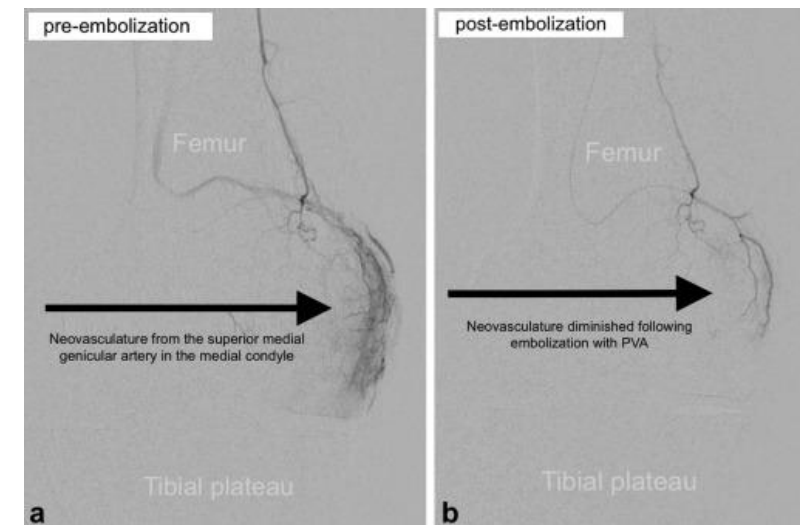
Pain from KOA



- Mechanical damage causes inflammation in the joint
- Pro-inflammatory factors are released that result in new blood vessel growth
- New sensory nerves grow along with the blood vessels and invade the joint space (neoinnervation)
- Movement of the joint then triggers pain signals *via* the new nerves
- One strategy to deal with the pain from KOA is to kill the new nerves to prevent the pain signalling

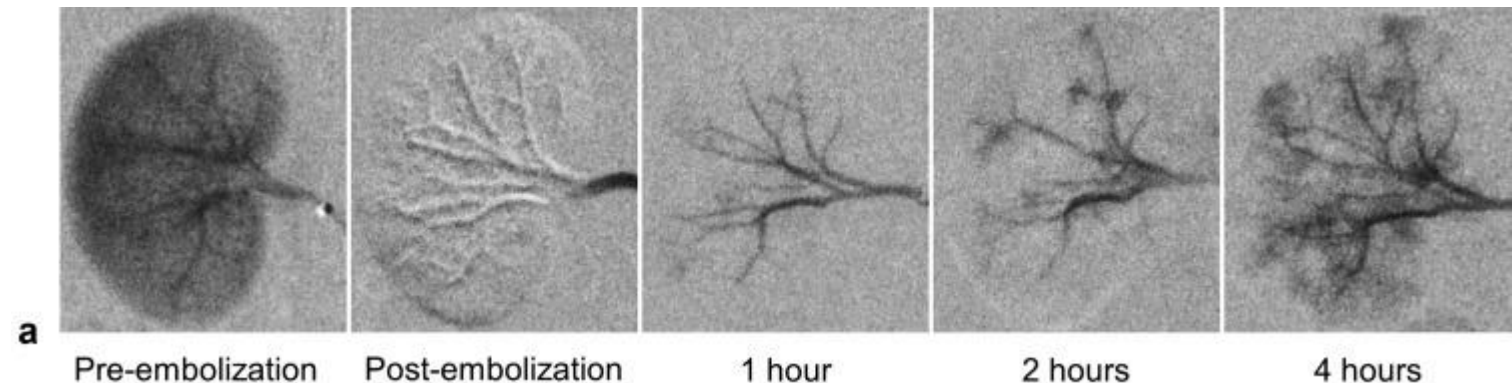
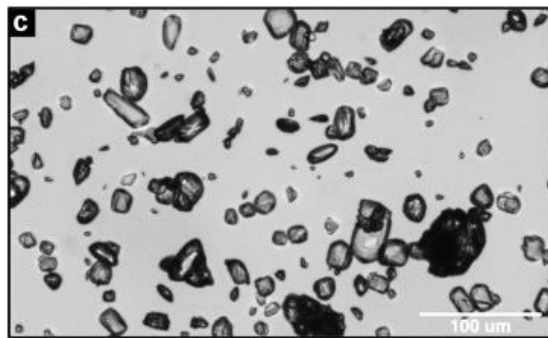
Genicular Artery Embolisation (GAE)

- Embolisation is the intentional blocking of a blood vessel for therapeutic purposes
- Performed by an Interventional Radiologist under image-guidance using minimally-invasive techniques
- Genicular Artery Embolisation (GAE) is a relatively new treatment where the arteries that feed the neovessels within the joint space are used to introduce particulate embolic agents
- These block the flow, create ischaemia and kill the blood vessels and associated nerves, reducing the pain signaling



Embolic Agents for GAE

- Many different embolic agents exist (both particulate and liquid) that are currently being assessed for their suitability to treat GAE
- The best clinical data are from Japan where the procedure is performed using a mixture of antibiotics (Imipenem:Cilastatin) that form particles when mixed with contrast agents
- The particles vary in size depending on how they are produced and dissolve away over a few hours, creating a transient ischaemia but allowing blood flow to recover (see below)
- This approach off-label and concerns over antibiotic use and allergic reactions means this agent is not suitable for long-term adoption in medical practice
- *An alternative device with similar properties is desired for mainstream GAE practice*



Key Design/Performance Requirements

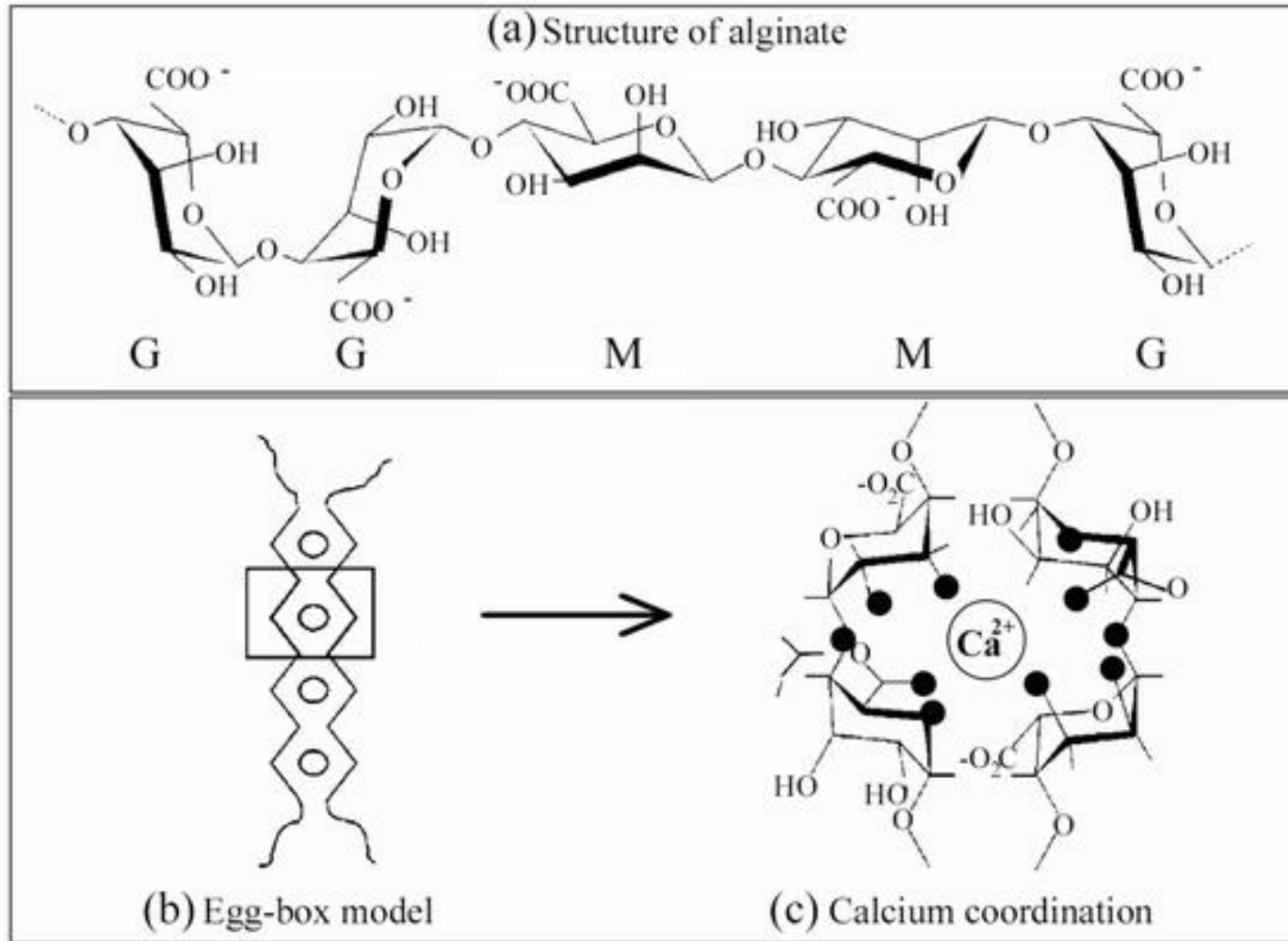
- Particulate embolic agents are favoured for GAE although Lipiodol liquid embolic is under investigation
- Your particles need to be deliverable through a microcatheter without blocking it and your research should have revealed that the ideal particle size for GAE is thought to be 200-300 microns
- Isobouyancy in contrast agent mixtures is desirable if possible (easily suspended with no or slow sedimentation to ensure smooth delivery)
- It is desirable (but not essential) to have visibility in suspension to aid in delivery and on Fluoroscope/CT (radiopaque) is a nice to have but would be challenging due to density
- Must be able to block an artery of a desired size at a particular location without migration to a non-target site
- Although permanent particles are being used for GAE, the current wisdom is that a degradable material would be preferable for added safety (as per IMP/CS)
- The particles need an appropriate mechanism and rate of degradation (1-3 hours and not dependent solely on slow hydrolytic processes) and be biocompatible/non-toxic when it breaks down (preferably metabolised or excreted)
- Be a sterile, off-the shelf product with a reasonable shelf-life (3 years)

SakuraBead® Temporary Embolic for GAE

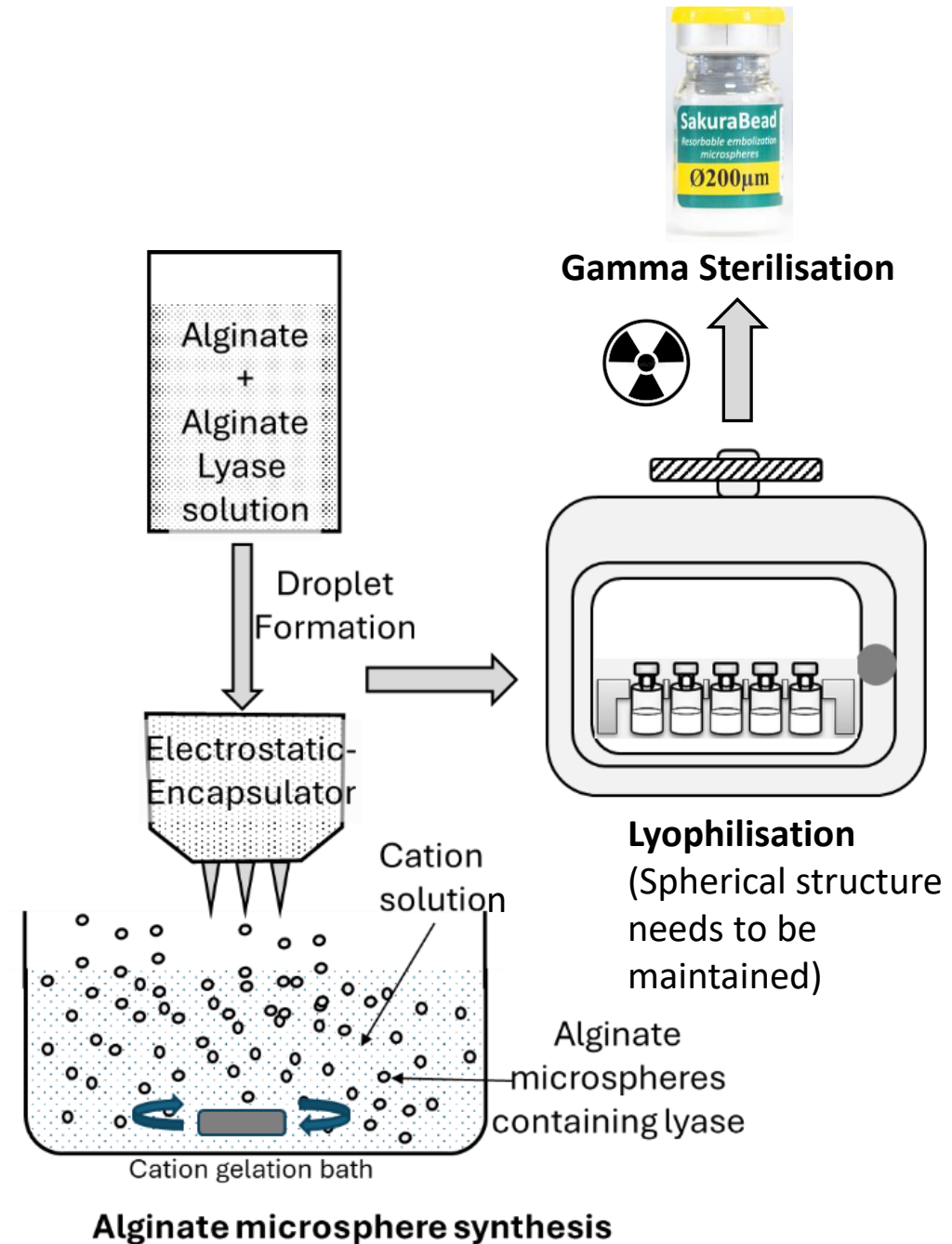
- Material Selection: Alginate sourced from seaweed
- Commonly-used biomaterial for biomedical applications which is highly biocompatible when very pure (proteins need to be removed)
- Hydrogel microspheres easily formed by calcium crosslinking using a variety of techniques (electrostatic droplet generator, jet cutter, microfluidics platforms)
- Contains alginate lyase enzyme embedded in the matrix to degrade the microspheres at a predetermined rate. Size controlled to $200 \pm 50\mu\text{m}$.
- Musculoskeletal embolization was pioneered in Japan. Cherry Blossom (**Sakura**) trees are synonymous with Japan & Japanese culture. The Sakura trees are known for the transience of their blossoms- hence SakuraBead.



Microsphere Manufacture



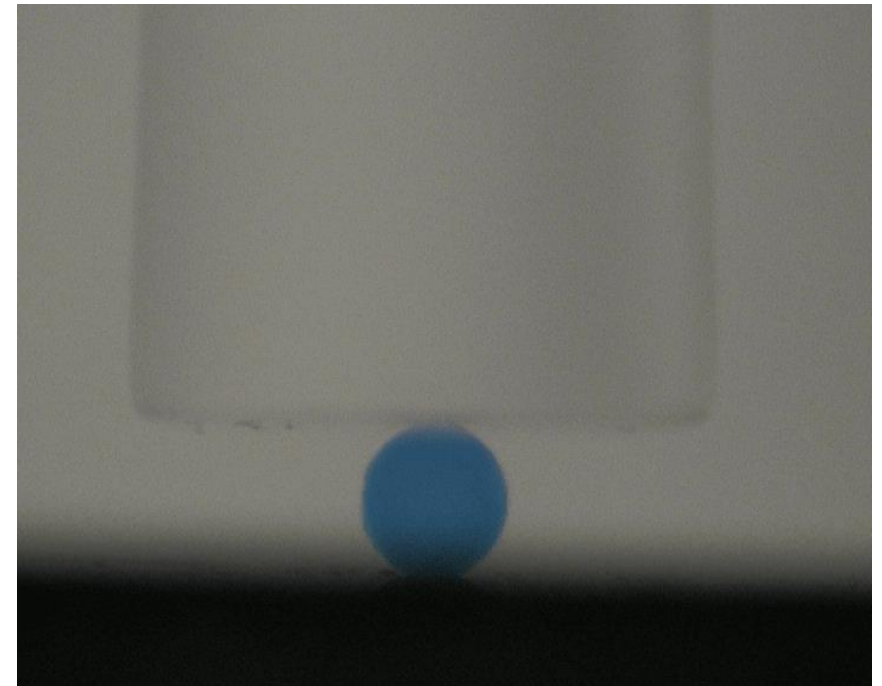
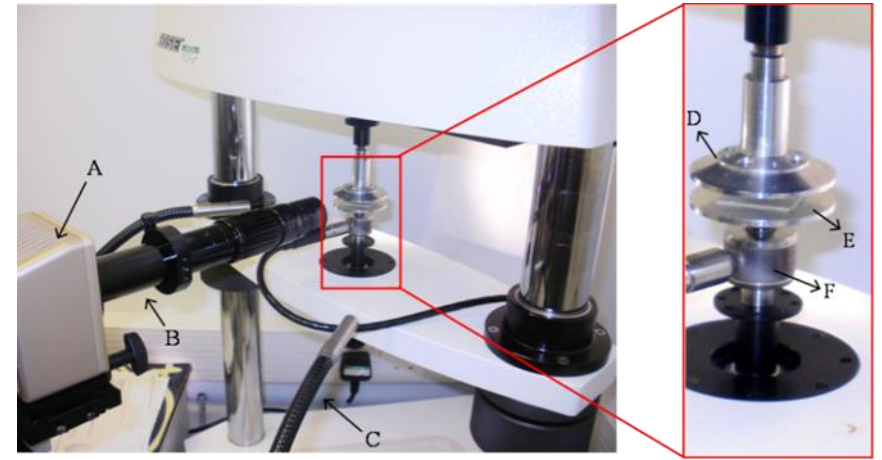
Alginate structure and crosslinking mechanism



Alginate microsphere synthesis

Compressibility and Isobouyancy

- Some hydrogel MS can be deformed significantly for delivery through small calibre microcatheters
- Material is mechanically robust and resistant to rupture under deformation as beads are viscoelastic and recover their shape rapidly once force is removed*
- But must resist migration once delivered into the target vessels
- Failure to recover shape post-delivery through microcatheters has led to clinically poor performance of other embolic microspheres**
- Hydrogels are also better for suspension in contrast medium without premature settling out which can make delivery difficult

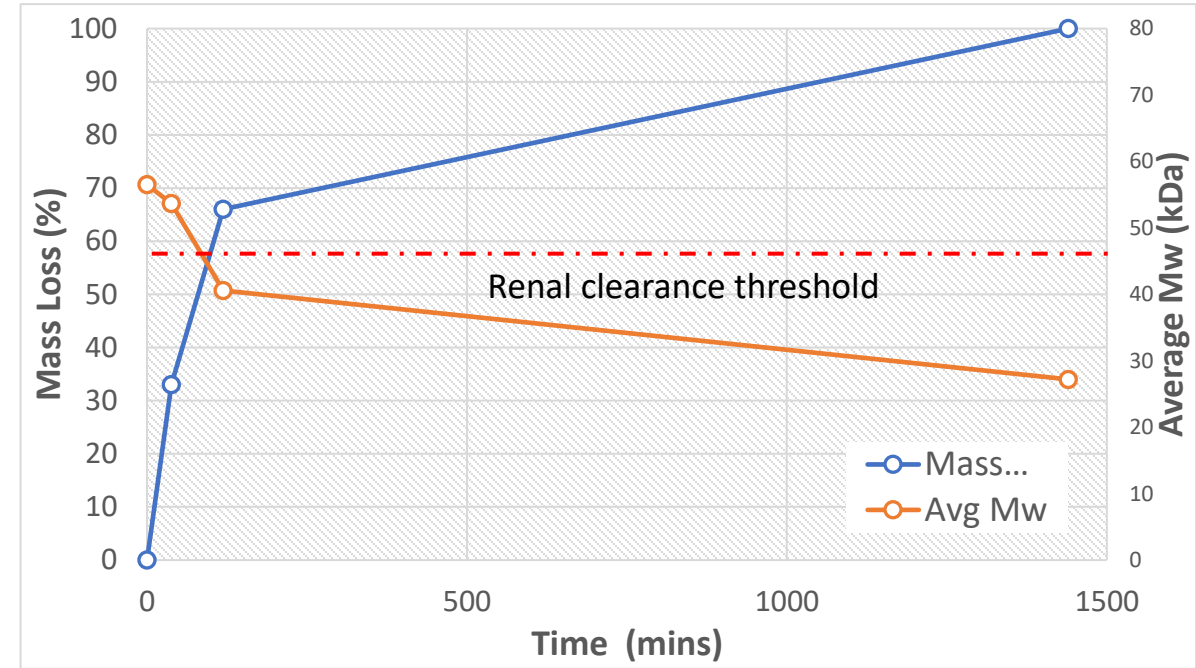


*Hidaka et al., *J Mech Behav Biomed Mater.* 2011;4(8):2161-7

**Spies et al., *J Vasc Interv Radiol.* 2005 Nov;16(11):1431-7

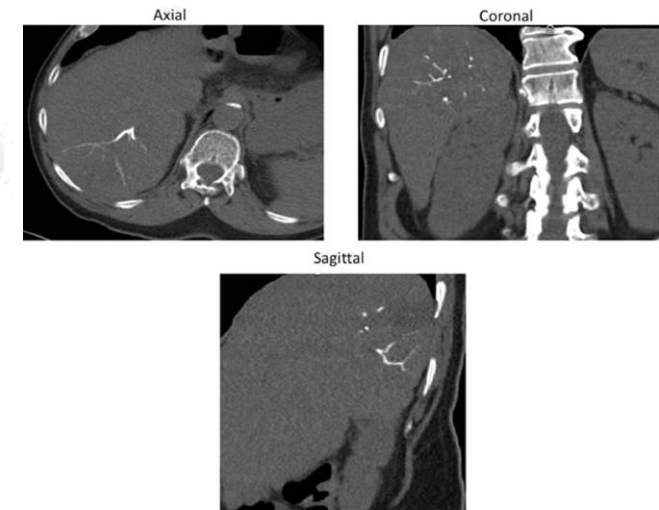
Degradation

- There are no enzymes in the body that will degrade alginate- the alginate lyase is specific for the device
- The rate of degradation can be tuned from minutes to weeks by controlling the alginate MW, M:G ratio and the amount of enzyme incorporated
- The alginate lyase cleaves the alginate chains reducing the molecular weight below that required for renal clearance
- Degradation rate is not dependent upon the patient's biological response nor the site of embolisation in the body (unlike gelatin and starch for instance)

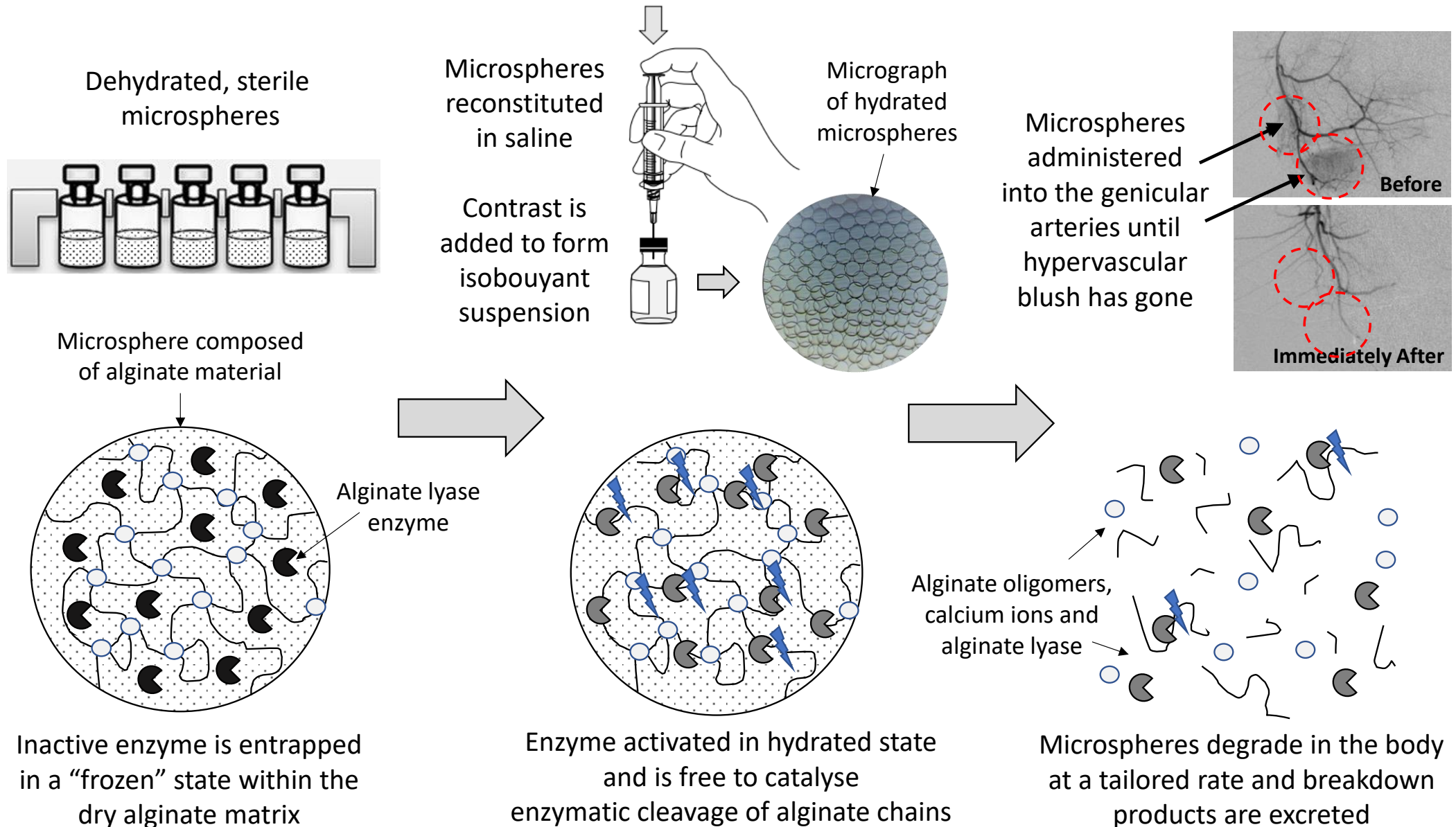


Colour and Radiopacity

- Permanent embolic particles are often dyed with a colouring agent so that they can be seen to form an even suspension for delivery
- This is more complex for a degradable embolic as the developer must demonstrate what happens to the dye in the body and that it is safe
- Radiopaque microspheres have been made by adding barium or tantalum particles, or by adding iodinated species to the MS polymer structure
- A lot is required to be visible (coating is insufficient) and the MS become very dense and difficult to deliver; in the GA they would be a challenge to visualise

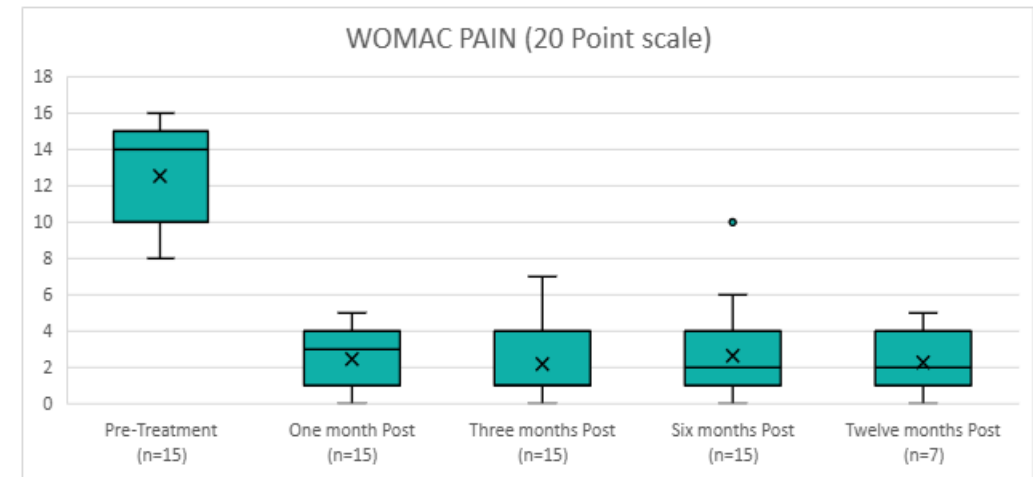
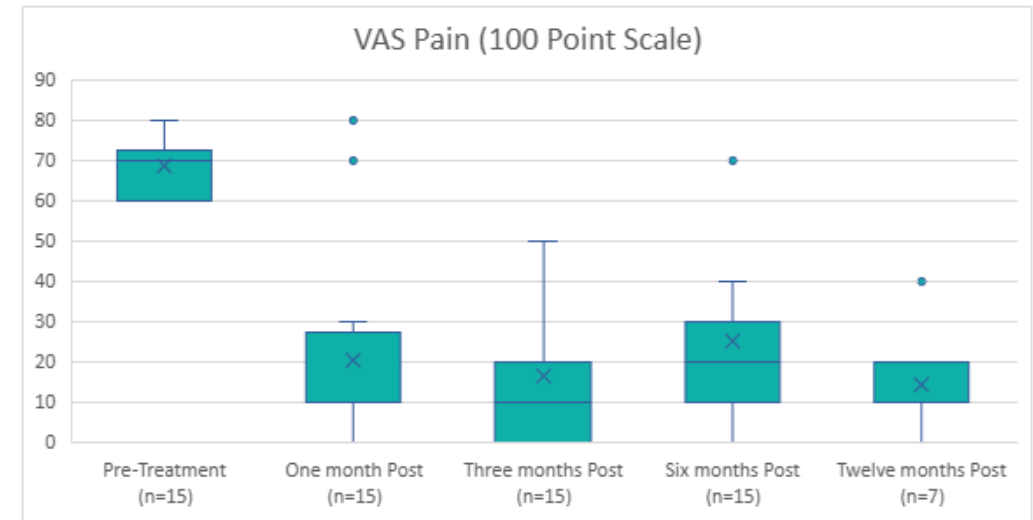


SakuraBead[®]: Preparation for Use



SakuraBead: First in Human Pain Relief Study

- 15 Patients treated with KL grade 2/3 KOA
- The average volume of SakuraBead suspension injected to reach the desired embolisation endpoint was 1.1 ml (a range of 0.4-1.9 mL)
- An average number of 2.4 genicular arteries were embolised per patient
- GAE was achieved in all the patients with 100% technical success
- Significant pain reductions seen post-procedure (14/15 patients) and sustained for at least 12 months



SakuraBead: A Combination Product

- Alginate microspheres are introduced into the genicular arteries and temporarily block the blood flow, inducing ischaemia which kills neovessels and neonerves in the joint space:
 - Primary mode of action: PHYSICAL EFFECT – DEVICE
- Alginate lyase enzyme inside the beads is activated once reconstituted and acts solely on the beads to degrade them into small fragments that can be excreted, independent of any degradative mechanism in the body. The presence and concentration of the enzyme control the rate and degree of degradation
 - Secondary action: BIOLOGICAL EFFECT – BIOLOGIC

FDA Classification: Device-Biologic Combination Product

I Hope You Enjoyed the Case Study

Any Questions?